Orbit Determination of Near-Earth Asteroid 2015 GF27 via the Method of Gauss

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ABSTRACT

The precise tracking of near-Earth asteroids (NEAs) is essential to survey populations and construct reliable impact-risk assessment. We determined the orbit of the NEA 2015 GF27 using three astrometric observations from the Las Cumbres Observatory network. Using recorded angular positions of the asteroid, combined with the reference Earth position vectors, a Python implementation of Gauss's method was used to determine to orbit of the asteroid. The solution converged using fourth-order f and g series, yielding orbital elements consistent with a near-Earth asteroid orbit. Comparison with JPL Horizons system showed agreement within approximately three percent for most orbital elements. We calculated a semi-major axis of 2.293190 AU (2.87% error), eccentricity of 0.44363 (2.82% error), inclination of 9.978° (1.56% error), longitude of the ascending node of 231.708° (0.13% error), argument of periapsis of 46.348° (1.69% error), and mean anomaly of 1.502° (16.71% error) relative to JPL Horizons values. The resulting orbital data could be submitted to larger databases and institutions, contributing to the scientific community's efforts to refine and improve the accuracy of asteroid orbit determinations.

1. INTRODUCTION

The population of small bodies orbiting near Earth has grown from a handful of cataloged objects in the mid-twentieth century to over thirty thousand known near-Earth asteroids (NEAs) today. This shift has been driven almost entirely by the maturation of wide-field survey astronomy. Early observations relied on astronomers to hand-draw the night sky and compare them with previous known charts to detect any major changes in position. In the late 1800s, Max Wolf pioneered the technique of taking long-exposure photographs and imprinting them onto photographic glass plates; stars would appear as distinct dots, while asteroids as streaks of light. By the early 1900s, astronomers were using the blink comparator, a tool that would rapidly move between two images, holding the background stars still so any asteroids would appear to jump back and forth (G. Wolfschmidt 2024).

The advent of dedicated electronic surveys in the 1990s, such as The Lincoln Near-Earth Asteroid Research (LINEAR) program, changed this picture substantially. By modeling survey completeness and detection efficiency, J. S. Stuart (2001) it produced one of the first statistically robust estimates of the size-frequency distribution of the NEA population. Subsequent work by W. F. Bottke et al. (2002) refined this picture further, debiasing the observed orbital and absolute-magnitude distribution of NEAs to account for the strong observational biases inherent in ground-based surveys.

A complementary, more recent debiasing of the NEA orbital and size distribution, was carried out by M. Granvik et al. (2018). They found systematic differences from the W. F. Bottke et al. (2002) model, particularly for orbits with very small perihelion distances – the distance of closest approach of an asteroid to the sun. Together, these efforts show that even the basic question of how many NEAs exist, and on what orbits, remains an active area of refinement that depends on having well-determined orbits for large numbers of individual objects.

The detection problem and the orbit determination problem are intimately linked. Population statistics can only be determined if individual detections can be converted into orbital elements reliable enough to permit recovery on subsequent nights, weeks, or years later. The fundamental algorithm for this was first developed by Gauss in the early nineteenth century, and uses three observations of an asteroid to solve iteratively for the position and velocity vectors of the object. From this, a complete set of orbital elements may be derived. While modern orbit determination pipelines typically employ differential correction and least-squares refinement against many observations (e.g., A. Milani & G. F. Gronchi 2010), the method of Gauss remains the standard technique for obtaining a preliminary orbit from a sparse initial observation set, and is still the algorithm of choice for first characterizing newly discovered objects.

The accuracy of any orbit determination is fundamentally limited by the astrometric precision of the observations and the time baseline over which the images are obtained. Survey instruments such as Pan-STARRS (J. L. Tonry 2011) and the wide-field infrared NEOWISE mission (A. Mainzer et al. 2011) demonstrate that systematic, repeated astrometric monitoring can both discover

new objects and tighten orbital uncertainties for known ones. Networks of robotic, geographically distributed telescopes, such as the Las Cumbres Observatory (LCO) network used in this work, extends this capability further. It enables rapid follow-up astrometry of newly announced or poorly characterized objects, something of increasing interest as survey discovery outpaces available resources.

Another practical motivation for this work is the assessment of collision risk. Although large (kilometer scale) NEA impacts are rare on human timescales, smaller objects in the tens-of-meters size range are far more numerous and arrive with comparatively little warning. The 2013 Chelyabinsk event, in which an approximately 20-meter object exploded in the atmosphere over Russia with an energy release equivalent to roughly 500 kilotons of TNT, injuring over a thousand people, is a stark reminder that even modest-sized, previously undetected objects can pose a real hazard (P. G. Brown et al. 2013). Because the consequences of an impact scale strongly, precise orbit determination is the single most important tool for converting a newly discovered point of light into an actionable assessment of risk.

This report applies the method of Gauss to determine an orbit for the near-Earth asteroid 2015 GF27, using three astrometric observations from the LCO telescope network. We aim to demonstrate classical orbit determination and characterize the resulting orbital elements and their uncertainties. Although the dataset used here is far smaller than that available to professionals survey pipelines, the underlying method is identical in spirit to the preliminary-orbit step used throughout the field, and this work serves as an illustration of how three observations and a few hours of calculation translate into a meaningful orbital solution.

2. OBSERVATION AND METHODS

2.1. *Instrumentation*

Astronomical images were captured at three different locations: the Cerro Tololo Inter-American Observatory, in Chile, the Haleakala Observatory, in Hawaii, and the McDonald Observatory, in Texas. The Las Cumbres Observatory (LCO) network determined the observation location on the basis of the position and brightness of the asteroid at different times. All observatories used a robotic reflecting telescope with a primary mirror diameter of 0.4 meters.

2.2. *Image Reduction*

Raw astronomical images are not directly suitable for photometric or astrometric analysis. Every CCD detector introduces systematic electronic and thermal signals that, if left uncorrected, distort the apparent brightness and position of sources on the detector. Although the images used in this work were delivered through the Las Cumbres Observatory (LCO) network, which applies an automated pre-reduction pipeline to all frames before delivery, we describe below the general procedure for CCD image reduction as implemented in `AstroImageJ`(K. A. Collins et al. 2017), the software used in this analysis.

2.2.1. *Calibration Frames*

Before processing the science images (referred to as "lights"), a set of calibration frames must be collected. These are:

Bias frames are zero-second exposures taken with the shutter closed that record the baseline electronic offset, and this offset must be removed first since it is present in all frames.

Dark frames are closed-shutter exposures that record thermally generated charge (dark current), which is then subtracted from the light images.

Flat frames are exposures of a uniformly illuminated surface that reveal pixel sensitivity variations and vignetting (when the center appears brighter); dividing by a flat field corrects this and makes the detector appear uniform.

Dark-flat frames are closed-shutter exposures taken at the same exposure time as the flat frames. They are required to remove the thermal and bias signal from the flat frames.

2.2.2. *Construction of Master Calibration Frames*

Multiple exposures are taken of each calibration type to suppress random noise. All master frames in this work use the *median* since averaging these frames can be skewed by cosmic-ray hits or other outlying pixel values.

The reduction sequence proceeds as follows. First, the individual bias frames are median-combined to produce a *master bias*, which is subsequently subtracted from all remaining frames.

The bias-corrected dark frames are median-combined to produce a *master dark*, which are then subtracted from the bias-corrected light frame.

A *master dark-flat* is formed by median-combining the bias-corrected dark-flat frames, which is subtracted from the bias-corrected flat frames and median-combined to produce the *master flat*.

Finally, each bias and dark-corrected light frame are divided by the normalized master flat, correcting for the pixel-to-pixel response variations across the detector.

2.2.3. *Image Stacking*

After corrections have been applied, the reduced light images were aligned and median-stacked with respect to a set of 5 reference stars. This process improves the signal-to-noise ratio (SNR) of the image, which is scaled approximately $1/\sqrt{N}$ for N combined frames, while also reducing residual artifacts.

2.3. *Astrometry*

The calibrated images were plate-solved in ASTAP, which assigned celestial coordinates to each pixel in the image. We can reference the header to find the location and date our asteroid was captured, and then use JPL Horizon System to get the approximate location coordinates of the asteroid during a period of time. The asteroid was identified by switching between multiple images and locating which object had moved relative to the background stars. After, we referenced the `SAOImageDS9`(Smithsonian Astrophysical Observatory 2000) to find the astreoid in `AstroImageJ`(K. A. Collins et al. 2017), and used the aperture measurement tool to circle it.

2.4. *Photometry*

For photometry, stars from the UCAC4 catalog were selected in SAOImage DS9 using the criteria

$$V_{\text{mag}} < 14 \qquad \text{and} \qquad V_{\text{mag}} \neq 0$$

These stars served as photometric reference stars because their apparent magnitudes are known. AstroImageJ's aperture photometry tool was used to measure the reference stars, and was adjusted to enclose the light from the source while minimizing background contamination. For each reference star, the measured right ascension, declination, and Source-Sky counts were recorded from AstroImageJ, while the catalog right ascension, declination, and V-band magnitude were recorded from the UCAC4 catalog in DS9. A calibration curve was then generated by fitting a line to the relationship between V_{mag} and $\log(\text{Source} - \text{Sky})$, with an expected slope of approximately -2.5. The quality of the calibration was evaluated by comparing the fitted slope to this expected value. Finally, the asteroid's measured Source-Sky value was substituted into the calibration equation to determine its apparent V-band magnitude.

2.5. *Orbit Determination Through the Method of Gauss*

Python implementation was utilized to iteratively solve for the straight-line position and relative velocity of the asteroid using the Method of Gauss. We referenced JPL Horizon system for the Earth-Sun vector, and found the ρ vector direction from the right ascension and declination of our observations. In addition, the difference in recorded time intervals from the three distinct observations, expressed as tau, was used as an initial guess for the Gauss coefficients. From these inputs, we iterated through the fourth order Taylor series for Gauss's f and g series, with a tolerance of 10^{-10}.

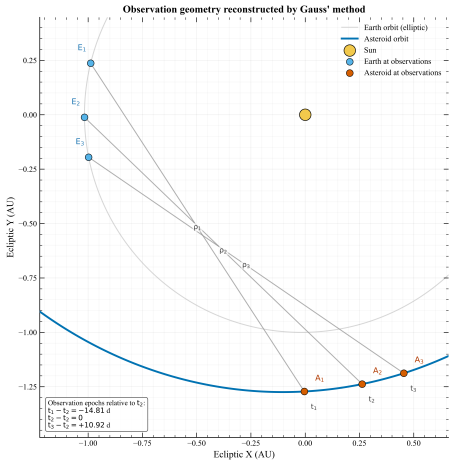


Figure 1. Observation geometry reconstructed by the Method of Gauss. Earth and asteroid positions are shown at each of the three observation epochs, with $t_1 - t_2 = -14.81$ d and $t_3 - t_2 = +10.92$ d.

3. RESULTS AND DISCUSSION

3.1. *Computed orbit*

The geometry of the three observations, reconstructed via the Method of Gauss, is shown in Figure 1, and the resulting three-dimensional orbit is shown in Figure 2. Figure 3 compares the recovered orbit to the JPL Horizons reference orbit; the two are visually near-identical near the observation epochs but diverge slightly elsewhere along the orbit.

Table 1 gives the computed orbital elements alongside the JPL Horizons reference values and the percentage difference between them, ranging from 0.13% in longitude of the ascending node to 16.71% in mean anomaly.

3.2. *Monte Carlo error propagation*

A percentage difference alone does not indicate whether the disagreement is consistent with astrometric measurement noise. To accurately quantify this measurement noise, empirical residuals were derived from the background star field in each observation. For each of the three epochs, a sample of 20 reference stars was selected. The centroids of these stars were measured to determine their observed right ascension (α) and declination (δ), which were then differenced against their known catalog coordinates. The standard deviation of

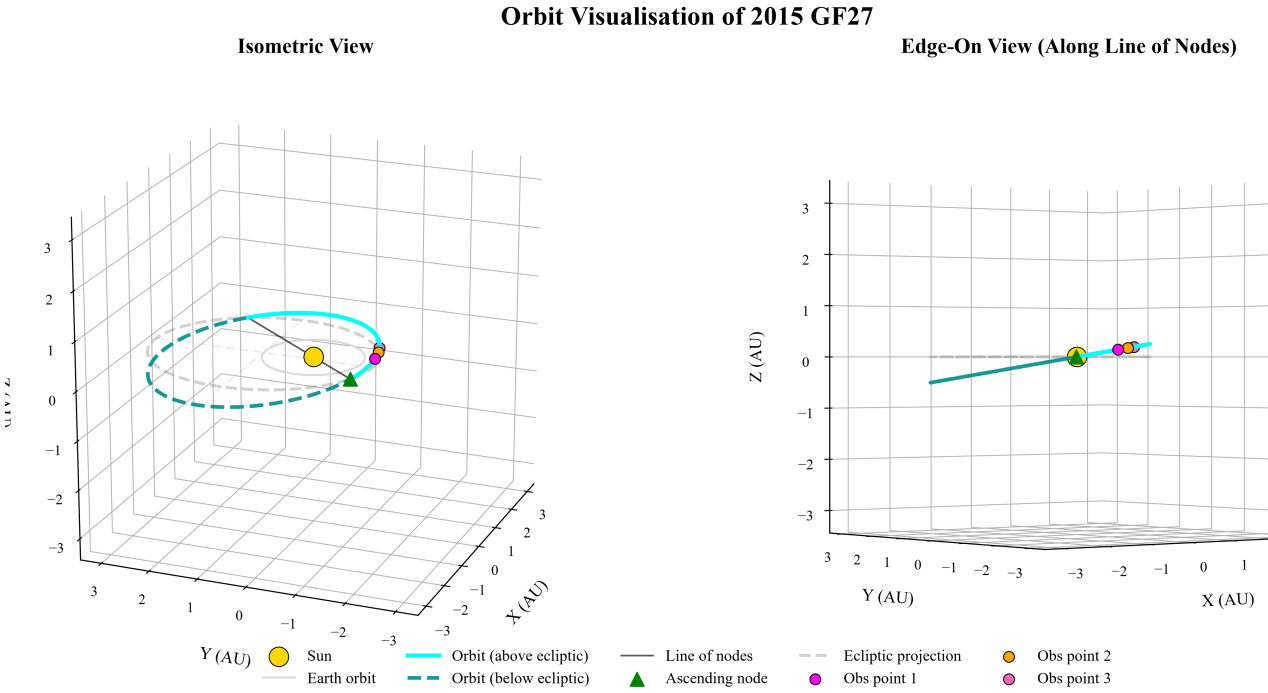


Figure 2. Three-dimensional orbital reconstruction of asteroid 2015 GF27 based on three observation epochs. The isometric view (left) details the asteroid's path relative to Earth's orbit, highlighting the perihelion and observation points. The edge-on view (right) looks directly down the line of nodes to illustrate the asteroid's true inclination relative to the ecliptic plane.

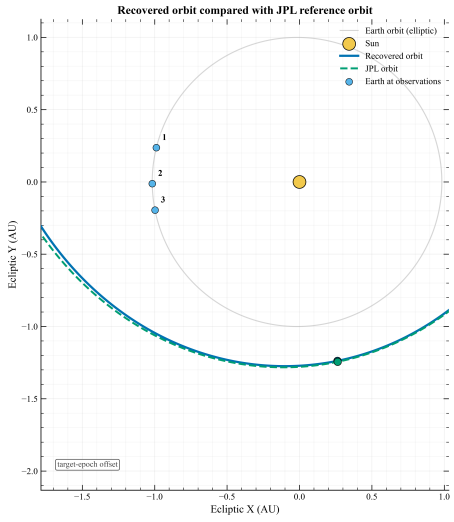


Figure 3. Recovered orbit compared with the JPL Horizons reference orbit.

these coordinate differences served as the astrometric uncertainty (σ_α and σ_δ) for that specific image, as summarized in Table 2.

A Monte Carlo (MC) error propagation was then performed to establish the baseline of this noise: the right ascension and declination of the asteroid's observations were perturbed by sampling normally from zero-mean distributions scaled by these measured uncertainties, and the full pipeline was rerun on each perturbed set. The resulting distributions are shown in Figure 4.

Element	Computed value $\pm\sigma_{MC}$	JPL (true) value	% difference
a (AU)	2.293190 ±0.00522	2.361019	2.87
e	0.44363 ±0.00104	0.45652	2.82
i (°)	9.978 ±0.0158	10.136	1.56
Ω (°)	231.708 ±0.00931	231.401	0.13
ω (°)	46.348 ±0.0211	47.143	1.69
M (°)	1.502 ±0.0042	1.287	16.71

Table 1. Computed orbital elements, with computed monte carlo uncertainties, for 2015 GF27 compared to the JPL Horizons reference values, with percentage difference.

Observation	σ_α (°)	σ_δ (°)
1	0.000 200	0.001 877
2	0.000 103	0.000 154
3	0.000 103	0.000 154

Table 2. Empirical astrometric uncertainties derived from the standard deviation of 20 reference star coordinate residuals for each observation epoch.

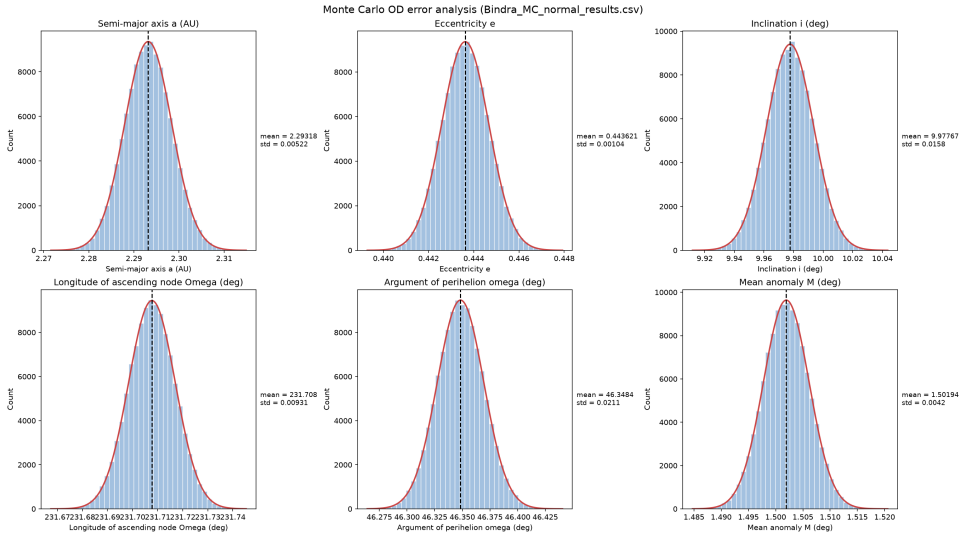


Figure 4. Monte Carlo distributions of the six orbital elements obtained by perturbing the input RA/Dec within their measured astrometric uncertainty and rerunning the orbit determination pipeline.

The Monte Carlo standard deviations shown in Table 1 are all far smaller than the differences would allow if astrometric noise were the dominant error source. This indicates that the percentage differences in the table reflect a systematic bias in the orbit determination rather than measurement noise. The pipeline is precise, in that resampling the same observations within their noise level reproduces nearly identical elements every time, but that precision does not establish accuracy.

3.3. *Locating the systematic error*

The discrepancy is not uniform across elements. Ω, ω, and M carry percentage differences of 0.13%, 1.69%, and 16.71%, while a, e, and i have differences of 2.87%, 2.82%, and 1.56%.

With an observation arc of approximately 26 days ($\Delta t_1 \approx -14.8$ d, $\Delta t_3 \approx +10.9$ d), the larger error in Ω and ω is not attributable to an unusually short baseline in time; it is better explained by the specific orbital geometry of 2015 GF27 combined with how the Method of Gauss actually determines orientation. The longitude of the ascending node and argument of perihelion are not measured directly from the observations, but are derived from the direction of the orbital angular momentum vector $\mathbf{h} = \mathbf{r}_2 \times \mathbf{v}_2$: Ω comes from the in-plane orientation of $\mathbf{h}$ relative to the reference x-axis, and ω from the angle between the ascending node and perihelion within the orbital plane. With a moderate inclination of $i \approx 9.98°$, the orbital plane is only slightly tilted from the ecliptic, so the out-of-plane component of the asteroid's motion over the observed arc is small compared to its in-plane motion. Since Ω and ω are set almost entirely by that small out-of-plane signal, while a, e, and i are set primarily by the much larger in-plane motion, any noise or model error in the recovered $\mathbf{r}_2$ and $\mathbf{v}_2$ is amplified disproportionately in Ω and ω relative to the other elements.

The mean anomaly's 16.71% difference is best understood as a downstream consequence of the error in a rather than an independent error of its own. M is obtained through Kepler's equation, $M = E - e \sin E$, where E, the eccentric anomaly, depends on a and e. Because of this, small errors in a and e get propagated forward into the large error in M.

The smaller, more uniform errors in a, e, and i may be caused by a sampling bias – this data was computed with only 3 values whereas JPL values have been computed using 386 observations (JPL 2026). Another possible cause is the absence of planetary perturbations in the two-body model used here, since the JPL reference is generated from a fully perturbed n-body integration; over the arc used, this would manifest as a small, roughly uniform shift in the derived geometry rather than a large error concentrated in one element.

3.4. *Implications*

Because the dominant error is systematic, it will not average out with additional observations at the same epoch, and the Monte Carlo framework cannot detect or correct it, since it only propagates whatever input noise it is given. Further improvement would not come from more precise astrometry, since the Monte Carlo distributions in Figure 4 show that astrometric precision is already far tighter than the errors actually present. Given the arc already spans roughly 26 days, improvement is more likely to come from incorporating planetary perturbations into the dynamical model or from observations at a different orbital phase that better constrain

the line of nodes. More broadly, this illustrates that the internal consistency of a solution, as measured by resampling or Monte Carlo methods, is not by itself evidence of its accuracy, and reported orbital elements should be accompanied by an estimate of systematic model error alongside the formal statistical uncertainty.

4. CONCLUSION

In this study, we successfully applied the Method of Gauss to determine the preliminary orbit of the near-Earth asteroid 2015 GF27, utilizing three astrometric observations captured by the Las Cumbres Observatory network. The derived orbital elements demonstrated strong agreement with the established JPL Horizons reference orbit, particularly for the semi-major axis, eccentricity, and inclination, which matched the reference values to within an error margin of less than three percent.

While larger discrepancies were observed in the orientation elements and the mean anomaly, a Monte Carlo error propagation analysis confirmed that these deviations were not driven by astrometric measurement noise. Instead, they were the expected result of systematic limitations inherent to the methodology—specifically, the reliance on an unperturbed two-body dynamical model and the geometric constraints of a short, 26-day observational arc.

To build upon these findings, future research should focus on mitigating the identified systematic biases. More observations would be instrumental in further constraining the orbit. Further, transitioning the propagation framework from a two-body approximation to a fully perturbed n-body dynamical model—incorporating planetary gravity—will be crucial for resolving the cascaded errors in the mean anomaly.

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APPENDIX

A. ORBITAL ELEMENTS

This appendix summarizes the formulae used to compute the orbital elements from the position and velocity vectors $\mathbf{r}$, $\dot{\mathbf{r}}$ at the central observation epoch t_2. All vectors are expressed in heliocentric ecliptic coordinates, with $\mu \equiv G(M_\odot + m) \cong GM_\odot$. When time is measured in Gaussian days and distance in AU, $\mu = k^2 = 1$, where $k = 0.01720209895$.

SEMI-MAJOR AXIS, A

From the Vis-Viva equation,

$$a = \frac{1}{\left(\frac{2}{r} - \frac{v^2}{\mu}\right)}, \qquad \text{where } r = |\mathbf{r}|,\ v^2 = \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} \tag{A1}$$

ECCENTRICITY, E

From $h^2 = \mu a(1 - e^2)$,

$$e = \sqrt{1 - \frac{h^2}{\mu a}}, \qquad \text{where } \mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}} \tag{A2}$$

INCLINATION, I

$$\cos i = \frac{h_z}{h}, \qquad h = |\mathbf{h}| = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & z \\ v_x & v_y & v_z \end{vmatrix} \tag{A3}$$

LONGITUDE OF THE ASCENDING NODE, Ω

The projection of $\mathbf{h}$ onto the ecliptic plane has magnitude $h_{xy} = h \sin i$. Both components are needed for the quadrant check:

$$\sin \Omega = \frac{h_x}{h \sin i}, \qquad \cos \Omega = -\frac{h_y}{h \sin i} \tag{A4}$$

ARGUMENT OF PERIHELION, ω

ω is computed as $\omega = U - \nu$, where U is the angle from the ascending node to $\mathbf{r}$, and ν is the true anomaly. Both are computed with quadrant checks:

$$\cos U = \frac{x \cos \Omega + y \sin \Omega}{r}, \qquad \sin U = \frac{z}{r \sin i} \tag{A5}$$

$$\cos \nu = \frac{1}{e}\left[\frac{a(1 - e^2)}{r} - 1\right], \qquad \sin \nu = \frac{a(1 - e^2)}{h\, e}\, \frac{\mathbf{r} \cdot \dot{\mathbf{r}}}{r} \tag{A6}$$

$$\omega = U - \nu \tag{A7}$$

MEAN ANOMALY, M

The eccentric anomaly E is obtained from r and ν, with both components needed for the quadrant check:

$$\cos E = \frac{ae + r \cos \nu}{a}, \qquad \sin E = \frac{r \sin \nu}{a\sqrt{1 - e^2}} \tag{A8}$$

The mean anomaly then follows from Kepler's Equation:

$$M = E - e \sin E \tag{A9}$$

To propagate M from t_2 to a target epoch t, the mean motion n is used, with $n = 2\pi/P = k/a^{3/2}$ (Kepler's Third Law):

$$M(t) = M(t_2) + n\,(t - t_2) \pmod{2\pi} \tag{A10}$$

B. THE METHOD OF GAUSS

This appendix summarizes the iterative Method of Gauss (MoG) used to determine the asteroid's position and velocity $\mathbf{r}_2$, $\dot{\mathbf{r}}_2$ at the central observation epoch t_2, from three RA/Dec observations. All calculations in this section are performed in equatorial coordinates, since the observed RA and Dec are equatorial; a rotation to ecliptic coordinates is required afterward, before the orbital elements themselves are computed.

SETUP

Given three observations $(\alpha_1, \delta_1, t_1)$, $(\alpha_2, \delta_2, t_2)$, $(\alpha_3, \delta_3, t_3)$ and the corresponding Sun vectors $\mathbf{R}_1$, $\mathbf{R}_2$, $\mathbf{R}_3$, the topocentric unit vectors are

$$\hat{\boldsymbol{\rho}}_i = (\cos\alpha_i \cos\delta_i,\ \sin\alpha_i \cos\delta_i,\ \sin\delta_i) \tag{B11}$$

Since $\mathbf{r}_2$ lies in the orbital plane along with $\mathbf{r}_1$ and $\mathbf{r}_3$, it can be written as

$$\mathbf{r}_2 = a_1\mathbf{r}_1 + a_3\mathbf{r}_3 \tag{B12}$$

for scalars a_1, a_3 to be determined. Working in Gaussian days ($\tau = k\,t$, with $k = 0.01720209895$, so that $\mu = k^2 = 1$), the Gaussian time intervals are

$$\tau_1 = k(t_1 - t_2), \qquad \tau_3 = k(t_3 - t_2), \qquad \tau_0 = k(t_3 - t_1) \tag{B13}$$

with $\tau_1 < 0$. Initial ("crummy") estimates of a_1, a_3 follow from Kepler's second law applied to the triangle approximation of the swept sectors:

$$a_1 \simeq \frac{\tau_3}{\tau_0}, \qquad a_3 \simeq \frac{-\tau_1}{\tau_0} \tag{B14}$$

SOLVING FOR THE ρ'S

Given a_1, a_3, the slant ranges ρ_1, ρ_2, ρ_3 are found from the vector equation

$$a_1\rho_1\hat{\boldsymbol{\rho}}_1 - \rho_2\hat{\boldsymbol{\rho}}_2 + a_3\rho_3\hat{\boldsymbol{\rho}}_3 = a_1\mathbf{R}_1 - \mathbf{R}_2 + a_3\mathbf{R}_3 \tag{B15}$$

which, via Cramer's Rule, gives closed-form expressions for each ρ_i in terms of triple products of the $\hat{\boldsymbol{\rho}}$'s and $\mathbf{R}$'s. The position vectors follow directly:

$$\mathbf{r}_i = \rho_i\hat{\boldsymbol{\rho}}_i - \mathbf{R}_i, \qquad i = 1, 2, 3 \tag{B16}$$

An initial estimate of $\dot{\mathbf{r}}_2$ is obtained by linear interpolation of the average velocities before and after t_2 (see Appendix A for the derivation):

$$\dot{\mathbf{r}}_2 = \frac{(t_3 - t_2)\,\mathbf{v}_{12} + (t_2 - t_1)\,\mathbf{v}_{23}}{t_3 - t_1}, \qquad \mathbf{v}_{12} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{t_2 - t_1}, \quad \mathbf{v}_{23} = \frac{\mathbf{r}_3 - \mathbf{r}_2}{t_3 - t_2} \tag{B17}$$

THE F AND G SERIES

Any position along the orbit can be written as a Taylor expansion about $\mathbf{r}_2$, $\dot{\mathbf{r}}_2$:

$$\mathbf{r}(\tau) = f(\tau)\,\mathbf{r}_2 + g(\tau)\,\dot{\mathbf{r}}_2 \tag{B18}$$

with f and g computed to fourth order in τ:

$$f(\tau) = 1 - \frac{\tau^2}{2r_2^3} + \frac{(\mathbf{r}_2\cdot\dot{\mathbf{r}}_2)}{2r_2^5}\,\tau^3 + \left(\frac{1}{24}\left[\frac{3v_2^2}{r_2^3} - \frac{1}{r_2^4}\right] - \frac{15}{r_2^5}\left[\frac{\mathbf{r}_2\cdot\dot{\mathbf{r}}_2}{r_2}\right]^2\right)\tau^4 \tag{B19}$$

$$g(\tau) = \tau - \frac{\tau^3}{6r_2^3} + \frac{(\mathbf{r}_2\cdot\dot{\mathbf{r}}_2)}{4r_2^5}\,\tau^4 \tag{B20}$$

Evaluating f, g at τ_1 and τ_3 gives two equations for $\mathbf{r}_1$, $\mathbf{r}_3$ in terms of $\mathbf{r}_2$, $\dot{\mathbf{r}}_2$; these are inverted to give refined values of both a_1, a_3 and $\mathbf{r}_2$, $\dot{\mathbf{r}}_2$:

$$a_1 = \frac{g_3}{f_1 g_3 - f_3 g_1}, \qquad a_3 = \frac{-g_1}{f_1 g_3 - f_3 g_1} \tag{B21}$$

$$\mathbf{r}_2 = \frac{g_3\mathbf{r}_1 - g_1\mathbf{r}_3}{f_1 g_3 - f_3 g_1}, \qquad \dot{\mathbf{r}}_2 = \frac{f_1\mathbf{r}_3 - f_3\mathbf{r}_1}{f_1 g_3 - f_3 g_1} \tag{B22}$$

ITERATIVE LOOP

Steps 1–5 are repeated until $\mathbf{r}_2$ converges:

1. Given a_1, a_3, solve for ρ_1, ρ_2, ρ_3.
2. Given the ρ's, compute $\mathbf{r}_1$, $\mathbf{r}_2$, $\mathbf{r}_3$ via $\mathbf{r}_i = \rho_i \hat{\boldsymbol{\rho}}_i - \mathbf{R}_i$.
3. Use $\mathbf{r}_1$, $\mathbf{r}_3$ to obtain refined $\mathbf{r}_2$, $\dot{\mathbf{r}}_2$.
4. Use the refined $\mathbf{r}_2$, $\dot{\mathbf{r}}_2$ to recompute f_1, f_3, g_1, g_3.
5. Use the refined f's and g's to recompute a_1, a_3, and return to step 1.

Convergence is declared when

$$\left| \frac{r_{2,n} - r_{2,n-1}}{r_{2,n}} \right| < 10^{-10} \tag{B23}$$